ENGINEERING RESEARCH CORPORATION LLC

Texas AI and Data-Center Infrastructure Reality Check

Distinguishing operating facilities, approved projects, credible build-outs, early proposals, and speculative queue entries in Texas's AI infrastructure boom — with scenario ranges, milestone-based confidence scoring, and verification-dated project records for the civic audiences making high-stakes decisions.

ERC Flagship Report ERC-RPT-2026-005

Version 1.0 — Public draft for review

Publication date: October 27, 2026

Engineering Research Corporation LLC

Prepared by the ERC Research Program

ERCOT filed the forecast in compliance with new PUCT requirements; it is built from utility-reported customer intentions, not from a prediction of what will be built. Source: ERCOT news release and PUCT Project 58777 filing, April 15, 2026 [1][2].

Document control and disclosures

Document control

Table i. Document-control record

Item	Detail
Report title	Texas AI and Data-Center Infrastructure Reality Check
Report identifier	ERC-RPT-2026-005 (persistent identifier; DOI planned as the ERC program matures)
Edition and version	Edition 1, Version 1.0 — Public draft for review
Publication date	October 27, 2026
Authors	ERC Research Program (named contributors listed in the final release)
Editors	Assigned at final release; independent technical review (grid planning and water planning) scheduled before publication
Publication status	Public draft for review. Project-level fieldwork and attributed interviews are scheduled for Edition 2; this edition verifies the anchor evidence and publishes the audit method.
Contact and canonical page	Engineering Research Corporation LLC. The canonical report page carries corrections, the dated project database, the data dictionary, and downloadable source tables; an interactive companion map is planned for Edition 2.
Suggested citation	Engineering Research Corporation LLC (2026). *Texas AI and Data-Center Infrastructure Reality Check*, ERC-RPT-2026-005, v1.0.
License and copyright	© 2026 Engineering Research Corporation LLC. Report text CC BY 4.0 on final release; project database and data dictionary under CC BY 4.0 and scripts under an OSI-approved license, unless a client contract requires otherwise.

Disclosures

Sponsor or client: None. ERC-initiated and self-funded. Future sponsorship will be disclosed on every distributed copy.

Conflicts of interest: ERC does not develop, own, or finance data centers, generation, or water infrastructure, and holds no position in any company discussed. ERC will not accept funding from parties to Texas data-center disputes for this report series.

Political neutrality: This report takes no position for or against data-center growth. It distinguishes what is known from what is claimed, presents distributional effects without averaging them away, and serves civic audiences on all sides of the question.

Right of reply and corrections: Because this report could influence public debate and investment, materially discussed organizations receive an opportunity to identify factual errors before final release — without editorial control over conclusions (Appendix F). A public correction log will be maintained.

Editorial independence: Findings are reported regardless of commercial or political effect.

Use of AI disclosure: This report was drafted with AI writing assistance under ERC staff direction. Every quantitative claim was verified against the cited primary source before

inclusion; no unverifiable project entry appears in the starter database.

Not legal or financial advice: Nothing here is investment advice, legal advice, or a recommendation for or against any project or bond issuance.

Abstract

Texas is at the center of the largest data-center build-out ever proposed, and the numbers circulating in public debate are mostly requests, not infrastructure. This report applies a project-confidence taxonomy — based on independently verifiable milestones including site control, permits, interconnection status, contracted power, financing, equipment orders, construction activity, announced customers, and expected energization — to distinguish five classes of projects: operating, approved-but-not-energized, credible with financing and interconnection evidence, early proposals, and speculative or duplicate queue entries. The verified anchor evidence is stark: ERCOT's preliminary 2026–2032 forecast projects 367,790 MW of demand by 2032 — more than four times the region's record peak of 85,508 MW — while ERCOT itself states the figure is likely higher than realized growth; the large-load interconnection queue stood near 445.8 GW of requests, roughly 87–89% data centers, against only 5.9 GW of observed energized large loads and 3.2 GW with approval to energize. Nationally, LBNL's 2025 update estimates data centers could reach 9.5–15.3% of US electricity by 2030 — a range, not a point. The report presents low, central, and high realization scenarios with stated assumptions rather than summing announced megawatts, analyzes electricity, transmission, dispatchable capacity, backup generation, water, land use, employment, tax incentives, curtailment, reliability, and cost allocation, and separates confirmed facts, company claims, forecast assumptions, stakeholder positions, and ERC conclusions throughout. Principal implication: Texas communities should plan against evidence-tiered scenarios and verification-dated project records — not press-release gigawatts.

Keywords

ERCOT; large load interconnection; data centers; Texas; long-term load forecast; water use; Batch Zero; Senate Bill 6; curtailment; cost allocation; scenario analysis; infrastructure planning

ONE-PAGE KEY FINDINGS — for local governments, water planners, journalists, and residents

The decision. Zoning, water, incentives, rate cases, and investments are being decided on data-center numbers of wildly different reliability. Which numbers can be trusted, and what should each audience do differently?

The verified numbers (all primary sources, verified October 27, 2026). Forecast: ERCOT's preliminary 2026–2032 LTLF projects ~278,003 MW by 2029 and ~367,790 MW by 2032 vs an 85,508 MW all-time peak — and ERCOT itself says it is likely higher than realized growth. Queue: ~445.8 GW of large-load interconnection requests (April 2026), ~87–89% data centers, +178 GW in one quarter — requests, not commitments. Reality: 5.9 GW observed energized large loads + 3.2 GW approved-not-operational = the only directly observed layer; 321 GW of requests have submitted no studies; 22 GW meet Section 9.5 requirements. Process: PUCT approved ERCOT's Batch Zero process (June 2026); the next forecast slipped to late 2026. National: LBNL's 2025 update puts data centers at 9.5–15.3% of US electricity by 2030 — a range, not a point. Water: Texas data centers ~25 billion gallons/year today (HARC); 3–9% of state water use possible by 2040 (UT Austin); TWDB received 18 of ~70 survey responses.

The reality-check ratio (ERC calculation). Observed energized large load (5.9 GW) divided by tracked requests (445.8 GW, same-date sources) = 1.3%; including approved-to-energize volume, 2.0%. The ratio is not a prediction — it quantifies how much of the public conversation rests on capacity that has not energized.

How to use this report. Classify every project by the five-class taxonomy (operating; approved; credible; early proposal; speculative/duplicate); quote numbers only with their date and status; plan against scenario ranges, not press-release gigawatts; and check every claim against its evidence tier.

Limitations. ERCOT's forecast methodology is in active revision; the queue contains duplicates and speculation by nature; water data is self-reported and sparse; incentive terms are often confidential; the project census completes in Edition 2 under the published rules.

Executive summary

Decision problem

Local governments, utilities, water planners, investors, and residents are making high-stakes decisions — zoning, water commitments, bond issuances, rate cases, investments — based on data-center numbers of wildly different reliability. Announced gigawatts, queue entries, and forecasts are being treated as equivalent to built infrastructure. This report provides the taxonomy, evidence rules, and verified anchors needed to tell them apart.

Key findings

1. The forecast and the queue are intention data, not infrastructure. ERCOT's preliminary 2026–2032 Long-Term Load Forecast (April 15, 2026, PUCT Project 58777) projects about 278,003 MW by 2029 and 367,790 MW by 2032, built from TDSP-reported medium and large load intentions — and ERCOT itself says it is likely higher than realized growth [1][2][3]. The Large Load Interconnection Queue reached roughly 445.8 GW of applications by April 2026, about 87–89% data centers, growing 178 GW in one quarter [3][4].

2. The observed base is two orders of magnitude smaller. 5.9 GW of observed energized large loads plus 3.2 GW approved-but-not-operational, per ERCOT's own tracking [3]. The maturity distribution inside the queue — 321 GW with no studies submitted, 93.7 GW under review, 22 GW meeting Section 9.5 requirements — confirms that most entries are early-stage.

3. Scenario ranges beat point forecasts. LBNL's *United States Data Center Energy Usage Report: 2025 Update* (published June 2026; DOI 10.71468/P1RP4F) estimates a reference case of 11.8% of US electricity by 2030 with a compounded uncertainty range of 9.5–15.3% (521–843 TWh) — deliberately a range [15]. ERC's scenarios (Section 7) apply the same discipline to Texas realization rates.

4. The state is rebuilding its methodology mid-stream. ERCOT proposed adjustments to its own preliminary forecast in May 2026 [6]; the PUCT approved ERCOT's Batch Zero interconnection process in June 2026, with ERCOT tracking more than 438,000 MW of large-load requests, about 89% data centers [7]; and the next Long-Term Load Forecast slipped to late 2026 [8]. Any single number quoted from this period should carry its filing date and status.

5. Water measurement is incomplete, and the credible studies bound rather than settle the question. HARC (January 2026) estimates about 25 billion gallons per year today, potentially 161 billion gallons (2.7% of state use) by 2030 across cooling and generation [17]; UT Austin (May 2026) projects 3–9% of state water use by 2040 depending on growth and cooling technology [18]; the Texas Tribune documented that the Texas Water Development Board received only 18 responses to about 70 data-center water surveys [19].

6. Policy is adapting in public. Texas Senate Bill 6 (2025) created large-load interconnection standards, cost-recovery contributions, mandatory curtailment capability for non-critical large loads, and backup-generation deployment authority [11][12]; its rules are

being implemented while the queue grows.

Confidence and limitations

High for the anchor numbers (all from ERCOT, PUCT, LBNL, or named research institutions, verified October 27, 2026). Moderate for scenario outcomes by design. The project-level database is a starter set of verified examples; Edition 2 completes fieldwork with the published taxonomy and verification rules. Major limitations: forecast methodology in revision; queue duplication by nature; sparse water data; confidential incentives; fast-aging construction status.

Reader guide

The **one-page key findings** (previous page) is the civic entry point. **Sections 1–3** establish purpose, questions, and the five-class taxonomy. **Section 4** documents the methodology, including the corrections and right-of-reply policy. **Section 5** presents the verified anchors with provenance. **Sections 6–7** organize findings and present the scenarios with competing interpretations and distributional effects. **Sections 8–10** cover limitations, stakeholder-specific recommendations, and the conclusion. **Appendices A–N** hold the taxonomy and scoring, the data dictionary, the full methodology, evidence-strength definitions, the interview protocol, the corrections policy, attributed stakeholder positions, the scenario arithmetic, the starter database, the ERCOT-documents primer, the civic FAQ, the verification-cadence calendar, the numbers card, and the accessibility conformance statement.

Contents

Document control and disclosures **2**

Document control . . . 2

Disclosures . . . 2

Abstract **3**

Keywords . . . 3

Executive summary **5**

Decision problem . . . 5

Key findings . . . 5

Confidence and limitations . . . 6

Reader guide . . . 6

Contents **7**

List of figures . . . 9

List of tables . . . 9

1. Purpose and decision context **11**

1.1 Why this report exists . . . 11

1.2 Primary audiences and decisions . . . 11

1.3 The governing qualification, restated . . . 11

2. Research questions **12**

3. Scope, definitions, and the project taxonomy **12**

3.1 Scope . . . 12

3.2 The five project classes (the report spine) . . . 12

3.3 Key definitions . . . 13

3.4 What counts as a duplicate: worked examples (ERC analysis) . . . 13

4. Methodology and evidence-selection protocol **14**

4.1 Sources . . . 14

4.2 Search and verification protocol . . . 14

4.3 Calculation methods . . . 14

4.4 Interview and stakeholder-engagement methods . . . 14

4.5 Independent technical review and right of reply . . . 15

4.6 Known limitations, stated up front . . . 15

5. Evidence landscape: the verified anchors 15

5.1 The forecast: preliminary by ERCOT's own description . . . 15

5.2 The queue: requests, maturity, and the observed base . . . 16

5.3 Policy: SB 6 and the changing interconnection regime . . . 17

5.4 Water: the measurable baseline and its gaps . . . 17

Cooling technology and the water-intensity question . . . 17

5.5 National context: the range discipline . . . 18

5.6 How the numbers move: a filing register . . . 18

6. Findings by research question 19

6.1 RQ1 — The five-class composition of the boom . . . 19

6.2 RQ2 — What the forecast actually measures . . . 19

6.3 RQ3 — Queue realization . . . 19

6.4 RQ4 and RQ5 — Electricity-system and water implications . . . 20

6.5 RQ6 — Land, employment, and fiscal effects . . . 20

6.6 RQ7 — Curtailment, reliability, and cost allocation . . . 20

6.7 Siting patterns and land use (ERC interpretation from documented regional counts) . . . 21

6.8 RQ8 — What each audience should do differently . . . 21

7. Analysis: scenarios, competing interpretations, distributional effects 21

7.1 Three realization scenarios (Calculated result; arithmetic in Appendix H) . . . 21

7.2 Competing interpretations . . . 22

7.3 Reliability, curtailment, and cost allocation under SB 6 (deep dive) . . . 22

7.4 Fiscal framework: incentives, bonds, and who bears risk . . . 23

7.5 Distributional effects (who gets what) . . . 23

8. Risks, limitations, uncertainty, and data gaps 24

8.1 Quality assurance gates applied to this report . . . 24

9. Recommendations by stakeholder 25

9.1 Question checklist for communities and local governments . . . 25

10. Conclusion 26

References 27

Appendix A: Project taxonomy and confidence scoring 29

A.1 Milestone dimensions . . . 29

A.2 Class-assignment rules 29
A.3 De-duplication rules 29

Appendix B: Data dictionary for the project database 29

Appendix C: Methodology appendix (full protocol) 30

Appendix D: Evidence-strength definitions 30

Appendix E: Interview protocol 31

Appendix F: Corrections and right-of-reply policy 31

Appendix G: Stakeholder positions (attributed, neutral) 31

Appendix H: Scenario arithmetic 32

Appendix I: Starter project database (verified entries) 32

Appendix J: How to read ERCOT and PUCT documents (a civic primer) 33

Appendix K: Civic FAQ 34

Appendix M: The numbers card — every headline figure with its source and status 35

Appendix N: Accessibility conformance statement 36

Glossary and acronyms 37

Change log 37

Data and code statement 38

List of figures

Figure 1. The reality-check funnel: large-load capacity by maturity stage (gigawatts, ERCOT tracking, April 2026) 16

Figure 2. LBNL 2025 update: 2030 US data-center electricity scenarios (terawatt-hours and share of US electricity) 18

List of tables

Table i. Document-control record 2

Table 1. Primary audiences and the decisions this report supports 11

Table 2. Research questions and where they are addressed 12
Table 4. Key definitions 13
Table 5. The preliminary 2026–2032 Long-Term Load Forecast 15
Table 6. Large-load interconnection tracking (ERCOT primary sources) 16
Table 9. Three realization scenarios (illustrative arithmetic, not forecasts) 21
Table 13. Limitations, consequences, and mitigations 24
Table 14. Recommendations by stakeholder 25
Table 17. Stakeholder positions as publicly stated 31

1. Purpose and decision context

1.1 Why this report exists

Conclusion first. Texas's AI infrastructure debate is being conducted with numbers of wildly different reliability — forecasts, queue entries, press releases, and operating reality — presented as if they were the same kind of fact. This report gives civic audiences a way to tell them apart, and gives decision-makers scenario ranges and verification-dated project records instead of press-release gigawatts.

This is ERC's fifth flagship report and its first public-interest report. It applies ERC's core competency — determining whether a claim is actually supported by authoritative evidence — to the most consequential infrastructure question in Texas, with the most careful possible treatment of uncertain, rapidly changing, politically consequential data. The five-way separation of claims (confirmed fact, company claim, forecast assumption, stakeholder position, ERC conclusion) governs every table.

1.2 Primary audiences and decisions

Table 1. Primary audiences and the decisions this report supports

Audience	Decision this report supports
Local governments	Zoning, water and sewer commitments, incentive agreements, bond issuance
Utilities and grid planners	Transmission planning assumptions under the PUCT Triennial Reliability Assessment (Project 58777)
Infrastructure developers	Site selection, interconnection sequencing, realistic demand scenarios
Economic-development organizations	Incentive design that survives a queue correction
Investors and lenders	Which projects have financing-grade evidence
Water planners (TWDB, groundwater districts, river authorities)	Demand projections built on measured, not self-reported-only, data
Journalists and community organizations	Which claims to attribute to which evidence tier
Ratepayer advocates and Texas residents	Cost-allocation and reliability questions in PUCT proceedings

1.3 The governing qualification, restated

ERC conclusion. Every use of the 2026–2032 forecast numbers in public debate should carry ERCOT's own qualification: preliminary, built from utility-reported customer intentions, and likely higher than realized load growth [1][3]. This report enforces that rule on itself: the scenarios in Section 7 are built from the observed base and queue with explicit realization assumptions, not from the forecast point.

2. Research questions

Governing point. Eight questions structure the report; each is answered or explicitly left unresolved.

Table 2. Research questions and where they are addressed

#	Research question	Addressed in
RQ1	How much data-center load is operating, approved, credible, proposed, and speculative in Texas — as of the last verification date?	Sections 5, 6, Appendix I
RQ2	What does ERCOT's preliminary 2026–2032 forecast actually measure, and how should it be used?	Sections 5.1, 7
RQ3	What realization of the interconnection queue is consistent with observable milestones?	Section 7.1
RQ4	What are the electricity, transmission, dispatchable-capacity, and backup-generation implications under those scenarios?	Sections 6, 7.2
RQ5	What are the water implications, and how reliable is the underlying measurement?	Sections 5.4, 6.4
RQ6	What are the land-use, employment, and tax-incentive effects, and how are they distributed?	Sections 6.5–6.6, 7.3
RQ7	How do curtailment, reliability, and cost-allocation rules (SB 6, Batch Zero) change the picture?	Sections 5.3, 6.6
RQ8	What should each audience do differently given the evidence tiers?	Section 9

3. Scope, definitions, and the project taxonomy

3.1 Scope

In scope: data centers (AI, cloud, enterprise) and their supporting infrastructure in the ERCOT region and adjacent Texas markets; the electricity demand, transmission, generation, water, land, fiscal, and reliability dimensions of that build-out; and the forecasting and interconnection processes that produce the public numbers. **Out of scope:** cryptocurrency mining facilities except where they appear in ERCOT large-load categories; non-Texas markets except for national context; and any recommendation for or against specific projects.

3.2 The five project classes (the report spine)

Table 3. The five project classes and their qualifying evidence

Class	Name	Definition (verifiable milestones required)	Examples of qualifying evidence
C1	Operating	Energized and serving load, observed in grid or operational data	ERCOT Observed Energized classification; commissioning records; metered demand
C2	Approved, not energized	Interconnection agreement signed and approval to energize granted	ERCOT Approval-to-Energize status; executed agreements

Class	Name	Definition (verifiable milestones required)	Examples of qualifying evidence
C3	Credible	Site control plus at least two of: permits filed or issued, interconnection study advanced, contracted power, financing closed, major equipment orders	Permits; signed agreements; financing disclosures; equipment orders
C4	Early proposal	Announced intent without C3-level milestone evidence	Press releases; economic-development filings; land speculation reports
C5	Speculative or duplicate	Queue entries lacking distinct identity, site control, or credible progress; duplicates of other entries	Cross-check failures; unchanged status across filings

3.3 Key definitions

Table 4. Key definitions

Term	Definition
Large load	75 MW or above at a single site (SB 6 default threshold); ERCOT medium loads are 25–74.9 MW
Interconnection queue	Requests for large-load interconnection tracked by ERCOT; entry is not a commitment
Batch Zero	ERCOT’s PUCT-approved batch process for assessing large-load interconnection requests
TDSP	Transmission and distribution service provider; the utilities that report medium and large load intentions into the forecast
Observed Energized	ERCOT’s classification for large loads with metered energization
Realization rate	The fraction of requested capacity that actually energizes in a period (Section 7.1)
Section 9.5	The ERCOT planning-guide requirements large-load interconnection applicants must meet
Firm load shed	Mandatory load reduction during grid emergencies; SB 6 requires non-critical large loads to be curtailable
4CP	Four coincident peak; the transmission cost-allocation methodology under reevaluation
Evidence tier	Strong / Moderate / Weak / Claim-only, per Appendix D

3.4 What counts as a duplicate: worked examples (ERC analysis)

De-duplication is where queue counts most often go wrong. Three worked cases from the rule set in Appendix A:

Case 1 — same site, two filings: a developer files a 400 MW request, then resubmits at 600 MW after a design change. Both entries share the parcel. Rule: the higher-evidence entry absorbs the lower; the 600 MW entry survives, and the merge is logged. Headline queue counts that skip this step double-count 200 MW.

Case 2 — parent entity, multiple counties: one company files in three counties. Rule: three distinct entries (different sites), each scored on its own milestones — but portfolio-level summaries must disclose that one balance sheet stands behind all three, which is material to financing-based confidence.

Case 3 — the whispered duplicate: two differently named projects at the same interchange, filed by entities sharing a registered agent, neither with site control. Rule: flagged as suspected C5 duplicates; both remain in the database with the flag visible, because a merger decision must be evidence-based and reversible.

4. Methodology and evidence-selection protocol

Governing point. The methodology is designed for contested, fast-moving data: primary sources only, verification dates on every fact, milestone-based confidence scoring, and explicit separation of facts from claims.

4.1 Sources

ERCOT: news releases; PUCT filings (Project 58777, Triennial Reliability Assessment); Board materials; ERCOT Monthly; market notices; Large Load Interconnection Queue tracking. PUCT: project filings and SB 6 rulemakings. Legislature: SB 6 text and analyses. Federal: LBNL data-center energy reports (2024 report; 2025 update) and DOE publications. Water: TWDB survey practices; groundwater district records; HARC and UT Austin studies; public permit records. Company disclosures: SEC filings and official releases — always typed as Company claims unless independently corroborated. Interviews: conducted under the Appendix E protocol, scheduled for Edition 2.

4.2 Search and verification protocol

Search dates for this edition: October 20–27, 2026; every cited page was opened and checked against the exact claim, with access dates in References. **Verification dating:** every project entry and time-sensitive number carries its last verification date and filing status (preliminary, proposed, approved). **Inclusion:** primary sources for quantities; named-authority publications for studies; company claims only when corroborated or explicitly typed. **Exclusion:** aggregators as primary sources; unverifiable announcements; social-media figures. **Duplicates and conflicts:** queue entries are de-duplicated by site, entity, and request identity before counting (Appendix A); conflicting official numbers are both reported with dates and status.

4.3 Calculation methods

Scenario arithmetic is deliberately simple and fully shown (Appendix H): realization-rate applications to the verified queue and observed base, with every assumption stated. No hidden weighting and no proprietary model. National figures are quoted from LBNL with their scenario structure intact.

4.4 Interview and stakeholder-engagement methods

Edition 2 conducts attributed interviews under the Appendix E protocol: on-the-record or on-background at the interviewee's choice, quotes checked for accuracy, statements typed as Confirmed fact, Company claim, or Stakeholder position, and conflicting accounts presented side by side. This edition makes no interview-based claims.

4.5 Independent technical review and right of reply

Before final release: at least one independent technical reviewer with grid-planning expertise and one with water-planning expertise review methods and conclusions; every materially discussed organization receives the factual-error opportunity under Appendix F — verified factual errors are corrected, while interpretations and conclusions remain ERC's alone; and all corrections appear in a public correction log. This review structure exists because the report could influence public debate and investment.

4.6 Known limitations, stated up front

The forecast methodology is under active revision at ERCOT and the PUCT; queue composition changes monthly; water data is largely self-reported; incentive agreements are often confidential; and the starter database covers only a verification-feasible subset of projects. Section 8 maps each limitation to a mitigation.

5. Evidence landscape: the verified anchors

Lead point. Every number in this section is from an official ERCOT, PUCT, legislative, federal-laboratory, or named-institution source, verified October 27, 2026, with filing date and status attached. These are the load-bearing facts of the Texas debate.

5.1 The forecast: preliminary by ERCOT's own description

Table 5. The preliminary 2026–2032 Long-Term Load Forecast

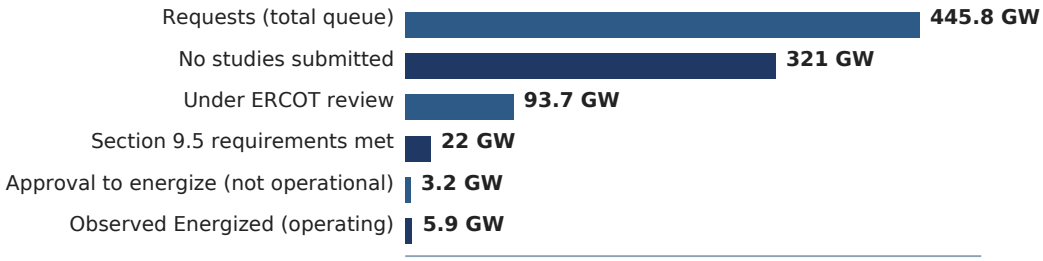
Item	Value	Status and source
Projected demand, 2029	~278,003 MW	Preliminary LTLF filed with PUCT April 15, 2026 (Project 58777) [2]
Projected demand, 2032	~367,790 MW	Same filing [1]
ERCOT all-time peak demand	85,508 MW	Same filing (observed record)
ERCOT's own caveat	Preliminary; likely higher than realized load growth; methodology refinement with PUCT and stakeholders underway	ERCOT Monthly, April 2026 [3]
Summer 2026 peak projection	90,500–98,000 MW	ERCOT comments [10]
Forecast inputs	Base economic forecast plus TDSP-provided information on medium (25–74.9 MW) and large (75+ MW) loads	ERCOT release [1]

5.2 The queue: requests, maturity, and the observed base

Table 6. Large-load interconnection tracking (ERCOT primary sources)

Item	Value	As of	Source
Large Load Interconnection requests	~410 GW (rising to 445.8 GW in ERCOT Monthly tracking)	Mar 26 / Apr 2026	Board Item 9 [4]; ERCOT Monthly [3]
Share that is data centers	~87% (Board) to ~89% (June release)	2026	Board Item 9 [4]; ERCOT June release [7]
Growth in one quarter	+178 GW since end of 2025; 198 new requests in Q1 2026	Q1 2026	Board Item 9 [4]
No studies submitted	321 GW	April 2026	ERCOT Monthly [3]
Under ERCOT review	93.7 GW	April 2026	ERCOT Monthly [3]
Section 9.5 requirements met	22 GW	April 2026	ERCOT Monthly [3]
Observed Energized large loads	5.9 GW	April 2026	ERCOT Monthly [3]
Approval to energize, not operational	3.2 GW	April 2026	ERCOT Monthly [3]
Batch Zero approved	PUCT approved ERCOT's Batch Zero process; more than 438,000 MW tracked, ~89% data centers	June 18, 2026	ERCOT release [7]
Next forecast	New LTLF not expected until mid-to-late December 2026; NERC LTRA uses a hybrid forecast	September 2026	Board Item 8 [8]

Figure 1. The reality-check funnel: large-load capacity by maturity stage (gigawatts, ERCOT tracking, April 2026)



Source: ERCOT Monthly, April 2026 [3] and Board Item 9 [4]. Units: gigawatts of large-load capacity (75 MW and above per site). The stages are ERCOT's own status categories; totals may include overlaps, so the funnel is a maturity snapshot, not a strict partition.

Reading Figure 1: 321 GW of the queue has not yet submitted studies — the speculative mass of the boom — while the directly observed layer (5.9 GW operating plus 3.2 GW approved) is roughly one-fiftieth of the total queue.

The reality-check ratio (Calculated result; formula shown). Observed energized large load (5.9 GW) divided by tracked large-load requests (445.8 GW, same-date sources) =

1.3%. Adding approval-to-energize volume (3.2 GW), the observed-plus-approved share of requests is **2.0%**. The ratio is not a prediction — realization rates vary by period and policy — but it quantifies how much of the public conversation rests on the roughly 98% that has not energized. Formula: observed ÷ requests; both figures from the same ERCOT Monthly edition so numerator and denominator share a date.

5.3 Policy: SB 6 and the changing interconnection regime

Texas Senate Bill 6 (89th Legislature, 2025; passed May 2025) established large-load interconnection standards and cost-recovery contributions (Utilities Code §37.0561); disclosure of on-site backup generation; mandatory capability to curtail during firm load shed for non-critical large loads; ERCOT authority to direct backup-generation deployment in emergencies; standardized interconnection processes; screening-study fees; and a reevaluation of the 4CP transmission-cost-allocation methodology [11][12][13]. ERCOT's implementation materials state the post-2026 inclusion standard: after 2026, only large loads with executed interconnection agreements satisfying the PUC Project 58481 requirements enter the Large Load forecast [9] — a structural improvement in forecast quality going forward.

5.4 Water: the measurable baseline and its gaps

HARC (January 2026): as of September 2025, Texas had more than 400 data-center facilities, concentrated in five regions (Dallas–Fort Worth 197; San Antonio 60; Houston 48); current consumption across cooling and generation is about 25 billion gallons per year, projected to as much as 161 billion gallons — 2.7% of state use — by 2030 under growth assumptions; at 77,965 MW of data-center demand, cooling alone would take 81–87 billion gallons annually [17].

UT Austin (May 2026): data centers are under 1% of state water use today and could reach 3–9% by 2040 depending on industry growth and cooling technology; the white paper recommends shared standards and integrated planning [18].

Measurement gap (Confirmed fact): TWDB sent about 70 water-use surveys to data centers in 2024 and received 18 responses; no state authority maintains a comprehensive data-center list [19][20].

National context: Ceres's *Water Behind the Watts* (August 2026) finds power generation dominates data-center-linked water use — about 3.4 trillion gallons of freshwater annually for electricity across the seven states analyzed, roughly twelve times the data centers' direct use — meaning grid composition, not just on-site cooling, drives the water footprint [22].

Company claims (typed as claims): the Stargate Abilene campus (Crusoe and related parties) reportedly expanded to 1.2 GW with a 940-acre site and an initial 8-million-gallon closed-loop fill from Abilene's system, whose average daily use is about 22 million gallons — utility statements corroborate the fill figure and system context [21].

Cooling technology and the water-intensity question

ERC analysis of documented engineering practice. The credible studies agree that cooling technology is a first-order driver of water outcomes: evaporative cooling consumes water continuously; closed-loop systems consume mainly an initial fill and occasional makeup, at some energy and capital cost; and air-cooled systems largely eliminate on-site water use at a power-efficiency penalty, which shifts water burden to generation (the Ceres finding). The Stargate Abilene project describes a closed-loop design with an initial 8-million-gallon fill [Company claim, corroborated in part by utility statements [21]]. The workbook records cooling-technology disclosure as a scored field for every project entry, because it is the single most decision-relevant water variable a local government can demand.

5.5 National context: the range discipline

LBNL's *United States Data Center Energy Usage Report: 2025 Update* (published June 2026; DOI 10.71468/P1RP4F) estimates data centers could account for 11.8% of total US electricity by 2030, with a compounded uncertainty range of 9.5–15.3% (521–843 TWh), derived from bottom-up simulation of facility types and locations; the 2024 report's range was 6.7–12.0% by 2028 (325–580 TWh), with 176 TWh consumed in 2023 (4.4% of US electricity) [14][15][16]. The methodological lesson for Texas: even the best-resourced federal-laboratory model publishes ranges, not points.

Figure 2. LBNL 2025 update: 2030 US data-center electricity scenarios (terawatt-hours and share of US electricity)



Source: LBNL, United States Data Center Energy Usage Report: 2025 Update (June 2026), DOI 10.71468/P1RP4F [15]. Units: TWh of US data-center electricity in 2030; shares are of total US electricity use in 2030. The range is the report's compounded uncertainty range.

5.6 How the numbers move: a filing register

Confirmed fact. The forecast and queue numbers cited in this report come from a sequence of filings and updates during 2026, each with its own status. The register below is the report's defense against quoting a superseded number; Edition 2 maintains it as a living table.

Table 7. 2026 ERCOT and PUCT filing register for large-load forecasting and interconnection

Date	Instrument	Content	Status as of October 2026
2026-04-13	ERCOT Board Item 9	Queue ~410 GW; ~87% data centers; +178 GW in one quarter; 198 new Q1 requests	Superseded by later totals

Date	Instrument	Content	Status as of October 2026
2026-04-15	ERCOT news release + PUCT Project 58777 filing	Preliminary LTLF 2026–2032: ~278,003 MW by 2029; ~367,790 MW by 2032; ERCOT caveat stated	Preliminary; adjustment proposed
2026-04-17	PUCT Open Meeting	Forecast discussed per new legislative requirements	Discussion item
2026-05-12	ERCOT Monthly (April)	Queue 445.8 GW; 5.9 GW observed energized; 3.2 GW approved-not-operational; maturity split (321 / 93.7 / 22 GW)	Current at publication
2026-05-18	ERCOT Market Notice M-B051826-01	Proposal to adjust the 2026 preliminary LTLF (two alternative methods, including Batch Zero-based)	Proposal
2026-06-18	ERCOT news release	PUCT approves Batch Zero process; more than 438,000 MW of large-load requests tracked, ~89% data centers	Approved process
2026-09-07	ERCOT Board Item 8	New LTLF not expected until mid-to-late December 2026; NERC LTRA uses hybrid forecast; Batch Zero good-cause exceptions approved	Current planning status

6. Findings by research question

Evidence-status note. Findings synthesize the Section 5 sources and carry the five-way separation — Confirmed fact, Company claim, Forecast assumption, Stakeholder position, ERC conclusion.

6.1 RQ1 — The five-class composition of the boom

ERC conclusion. As of the April–June 2026 ERCOT tracking: Class C1 (operating, observed) is about 5.9 GW; Class C2 (approved, not energized) about 3.2 GW; Classes C3–C5 make up the remainder of the roughly 445.8 GW of requests, with 321 GW having submitted no studies (predominantly C4 and C5) and 22 GW meeting Section 9.5 requirements (candidates for C3). The starter database (Appendix I) begins classifying named projects; Edition 2 completes the census under the published rules.

6.2 RQ2 — What the forecast actually measures

ERC conclusion. The preliminary Long-Term Load Forecast is a structured aggregation of customer intentions as reported by utilities — valuable for transmission planning under the Triennial Reliability Assessment, unsuitable as a prediction of built capacity. Three structural facts support this: ERCOT's own caveat [3]; the May 2026 adjustment proposal citing energization-date realism and the Batch Zero alternative [6]; and the post-2026 rule that only executed interconnection agreements enter the forecast [9].

6.3 RQ3 — Queue realization

ERC conclusion. Queue realization cannot be read off the queue. It depends on interconnection-study throughput (now batched), financing (not observable in queue data), and market demand for AI capacity. The evidence supports wide uncertainty, which is why Section 7 presents three scenarios with stated assumptions instead of one number.

6.4 RQ4 and RQ5 — Electricity-system and water implications

ERC conclusion (electricity). Under any plausible realization, Texas faces: transmission build-out concentrated in specific zones; dispatchable-capacity questions — the generation queue's 2,008 active requests totaling about 453,562 MW (storage 177,642 MW; solar 162,927 MW) are themselves mostly requests, not plants [9]; and growing use of the SB 6 curtailment and backup-generation tools during scarcity. Backup generation behind the meter raises cost-allocation and emissions questions the Legislature chose to regulate rather than assume away.

ERC conclusion (planning process, additional). The September 2026 Board update shows the practical consequences of the forecast delay: ERCOT submitted a hybrid peak load forecast (the April 2026 base econometric forecast plus the TSP-adjusted April 2025 Large Load forecast) to meet the NERC Long-Term Reliability Assessment deadline, and NERC approved use of its 15-percent default Reference Margin Level pending the assessment [8]. *Interpretation:* the state's reliability institutions are running on bridging measures until the methodology lands — one more reason every consumer of these numbers should track the filing register (Table 7) rather than memorize any single figure.

ERC conclusion (water). The water conversation currently runs on incomplete measurement: the credible studies (HARC, UT Austin) agree that growth is material and that cooling technology and grid composition dominate outcomes, while the state's own survey response rate (18 of about 70) shows the data foundation is weak [17][18][19]. Groundwater districts have begun asking the Legislature for clearer authority — a Stakeholder position, attributed in Appendix G [23].

6.5 RQ6 — Land, employment, and fiscal effects

ERC conclusion. Verified project footprints are large; the 940-acre Stargate Abilene site description is a Company claim corroborated by press descriptions [21], and county appraisal and permit records are the reliable source, scheduled for Edition 2 fieldwork. Employment effects split into a small permanent-operations workforce and a large temporary construction workforce. Incentive agreements (abatements and successor programs) are local and often confidential; announced figures are treated as Company claims and verified against local records in Edition 2.

6.6 RQ7 — Curtailment, reliability, and cost allocation

ERC conclusion. SB 6 changed the default answers to who pays and who flexes: large loads must be curtailable during firm load shed, contribute to interconnection-cost recovery, and — after 2026 — qualify for forecast inclusion only with executed agreements [11][12]. Batch Zero allocates available grid capacity across applicants in batches [7]. These rules materially

change the risk profile that early-2025 commentary assumed, but their effects depend on PUCT implementation details that remain in progress.

6.7 Siting patterns and land use (ERC interpretation from documented regional counts)

The documented regional distribution — of more than 400 Texas facilities as of September 2025, Dallas–Fort Worth had 197, San Antonio 60, Houston 48 — shows concentration in metropolitan peripheries with transmission access [17]. The next wave, however, is visibly rural: the Stargate Abilene campus sits outside Abilene [21], and Panhandle proposals sit over the Ogallala Aquifer, where community opposition has organized [19]. **ERC interpretation:** the siting frontier is moving from metro-adjacent to transmission-following rural sites, which changes which counties must prepare — and it means land-use evidence (parcels, options, appraisals) becomes the earliest reliable signal of where the boom actually lands, ahead of any grid data.

6.8 RQ8 — What each audience should do differently

Deferred to Section 9, with recommendations keyed to evidence tiers and dates.

7. Analysis: scenarios, competing interpretations, distributional effects

7.1 Three realization scenarios (Calculated result; arithmetic in Appendix H)

Assumption base (all from Section 5, same-date sources where possible): requests R = 445.8 GW (April 2026); observed energized L = 5.9 GW; approved-not-operational A = 3.2 GW; 2026 projected summer peak 90.5–98 GW; 2032 preliminary forecast 367.8 GW with ERCOT's caveat.

Table 9. Three realization scenarios (illustrative arithmetic, not forecasts)

Scenario	Realization assumption by 2032	Resulting large-load addition	Illustrative 2032 peak (arithmetic)	Status label
Low	5% of non-energized requests realize	~22 GW	~113–120 GW	Illustrative arithmetic, not a forecast
Central	15% realize	~66 GW	~157–164 GW	Illustrative arithmetic, not a forecast
High	30% realize	~132 GW	~223–230 GW	Illustrative arithmetic, not a forecast

(Formula: addition = r × (R − L − A); illustrative peak = 2026 peak + addition, ignoring ordinary growth. All three scenarios sit far below the 367.8 GW preliminary forecast — which is the point: the forecast's own inputs are intentions, and ERCOT says so. Historical realization rates for comparable interconnection queues should replace these illustrative rates in Edition 2; the rates above are chosen to span the plausible range, not derived from a

fitted model — an Estimate with stated assumptions. Full arithmetic in Appendix H.)

7.2 Competing interpretations

"The forecast is real demand." *Position held by* some market participants citing contracted timelines. *Evidence for:* TDSPs report named customers with engineering contacts; some C3-class projects have financing. *Evidence against:* ERCOT's caveat; the 321 GW no-studies mass; the 1.3% observed-to-requested ratio. **ERC conclusion:** the forecast is a planning envelope; treating it as a demand forecast is unsupported.

"The queue is mostly vapor." *Position held by* some skeptics. *Evidence for:* the maturity distribution and general interconnection attrition. *Evidence against:* 22 GW already meets Section 9.5 requirements; 3.2 GW holds approval to energize; several C3-class projects show verified construction. **ERC conclusion:** the queue is a wide distribution with a thin but real leading edge — exactly what the taxonomy exists to separate.

"Water is a non-issue" versus "water is the binding constraint." *Stakeholder positions on both sides, attributed in Appendix G.* **ERC conclusion:** the honest statement is measurement failure — the state's own data collection is too thin to adjudicate, and the credible studies bound the answer rather than settle it. Cooling technology and grid composition dominate outcomes.

7.3 Reliability, curtailment, and cost allocation under SB 6 (deep dive)

ERC analysis of the enacted framework. SB 6 changed three default answers that early-2025 commentary assumed away. First, **who flexes**: non-critical large loads connected at transmission must be equipped for curtailment during firm load shed, and loads with on-site backup generation can be directed by ERCOT (through their utility) to deploy it in emergencies — with such deployments counted as firm load shed for price-adjustment purposes [11][12]. Second, **who pays**: large-load customers contribute to interconnection-cost recovery under PUCT-standardized rules, pay screening-study fees, and the 4CP transmission cost-allocation methodology is under reevaluation [12][26]. Third, **who studies**: the PUCT standardized interconnection processes, and ERCOT moved to batch studies — the June 2026 Batch Zero approval allocates available grid capacity across applicants in batches rather than first-come-first-served [7].

Table 11. SB 6 and Batch Zero mechanisms and their evidence implications

Mechanism	What it changes	What evidence communities should track
Mandatory curtailment capability	Non-critical large loads shed during firm load-shed events	Utility attestations of curtailment equipment; exercised-in-emergency records
Backup-generation deployment authority	ERCOT can direct behind-the-meter generation during scarcity	Deployment logs; emissions and air-permit implications
Interconnection cost recovery	Large loads contribute to utility interconnection costs	PUCT filings; tariff provisions; actual charges over time

Mechanism	What it changes	What evidence communities should track
Screening-study fees	Study burden shifts toward applicants	Fee schedules; queue-processing statistics
Co-location review	Large loads netting against existing generators face review	PUCT review outcomes; generator-retirement interactions
Batch Zero	Interconnection assessed in capacity-limited batches	Batch study results; capacity allocations; deferral lists
4CP reevaluation	Transmission cost allocation may shift	PUCT rulemaking docket outcomes

Reliability framing (ERC conclusion). These mechanisms reduce — but do not eliminate — the reliability exposure of rapid large-load growth: curtailment spreads scarcity costs to the loads that cause them, but the underlying need for dispatchable capacity and transmission remains, and the generation queue that would supply it is itself mostly requests (about 453,562 MW of active requests, dominated by storage and solar) [9].

7.4 Fiscal framework: incentives, bonds, and who bears risk

ERC analysis framework. Fiscal effects arrive through four channels, each with different evidence quality: (1) **property-tax base** — large during construction, incentive-dependent after (abatement and successor-program terms are often confidential until agreements post; typed as Company claims until verified against local records); (2) **utility and ratepayer costs** — interconnection, transmission, and any reliability investments, allocated under SB 6 rules and 4CP outcomes; (3) **municipal service costs** — water, sewer, roads, and emergency services, which scale with construction phases; and (4) **incentive outlays** — grants, abatements, or chapter-agreement commitments. **ERC conclusion:** communities should model fiscal scenarios against the same realization tiers as electricity demand — an incentive agreement sized for a 1,200 MW campus behaves very differently if 300 MW energizes.

7.5 Distributional effects (who gets what)

Table 10. Distributional effects of the build-out

Effect	Concentrated on	Nature
Construction employment	Host counties; temporary	Large, finite, front-loaded
Permanent operations jobs	Host counties; small	Modest per MW; skill-specific
Property and appraisal base	Host counties	Large during construction; incentive-dependent after
Tax-incentive cost	Local taxpayers where abatements apply	Often confidential until agreements are posted
Grid costs and reliability risk	ERCOT-load customers broadly	Cost-allocation rules (SB 6, 4CP review) determine sharing
Water demand	Local aquifers and systems	Site-specific; measurement weak
Land use	Rural tracts near transmission	Visible, permanent

Effect	Concentrated on	Nature
Price and availability effects	Data-center customers versus other customers	Depends on curtailment and cost-allocation implementation

(No net score is computed; the report's public-interest duty is to make the distribution visible.)

8. Risks, limitations, uncertainty, and data gaps

Lead point. The most consequential limitation is that the numbers themselves are moving targets — the forecast methodology, the interconnection process, and the queue are all in active revision — which is why every figure here carries a date and status, and why the scenarios avoid dependence on any single filing.

Table 13. Limitations, consequences, and mitigations

Limitation	Consequence	Mitigation
ERCOT forecast methodology in active revision	Numbers move between filings	Filing date and status on every number; register of filings; scenarios avoid dependence on any single filing
Queue contains duplicates and speculation	Counts overstate distinct projects	De-duplication rules (Appendix A); the taxonomy separates C4 and C5
Water data self-reported and sparse	Water planning built on thin data	Report survey response rates; prefer measured studies; flag TWDB data gaps
Incentive terms confidential	Fiscal analysis incomplete	Type as Company claims; verify against local records in Edition 2
Construction status ages quickly	Stale database entries	Verification dates on every entry; refresh cycle stated
Politically contested environment	Pressure on framing	Neutral civic language; five-way claim separation; right of reply without editorial control; independent technical review
Interview evidence pending	Practitioner insight absent from this edition	Protocol published now (Appendix E); interviews in Edition 2
National context shifting	LBNL ranges move	Quote the report and its DOI; recheck at each edition

8.1 Quality assurance gates applied to this report

Editorial review. Title matches scope; the executive summary agrees with the body; every research question is answered or explicitly unresolved; conclusions stay within the evidence; the five-way claim separation is enforced in every table; neutral naming; acronyms defined at first use.

Source audit. Every link cited was opened and checked against the exact claim during preparation (access dates October 27, 2026). ERCOT, PUCT, legislative, LBNL/DOE, and named-institution sources are primary; news summaries are used for context and quoted utility statements with primary sources preferred; company-related descriptions are typed as claims.

9. Recommendations by stakeholder

Lead point (Recommendation). Plan against evidence tiers; quote numbers with their dates and status; and demand verification-dated project records — from developers, from your own staff, and from us.

Table 14. Recommendations by stakeholder

Stakeholder	Recommendation
Local governments	Classify every pipeline project by the Appendix A taxonomy before signing incentives or water commitments; require milestone evidence (site control, permits, financing) as a condition of further consideration; write incentive clawbacks keyed to energization, not announcement
Utilities and planners	Use scenario bands, not the preliminary forecast point, for distribution and transmission planning; track the PUCT's Batch Zero and forecast-adjustment proceedings; publish your own observed-energization statistics
Water planners	Treat TWDB data-center water figures as lower bounds given the 18-of-70 response rate; require cooling-technology disclosure in permits; coordinate with groundwater districts on cumulative-impact analysis
Developers and investors	Voluntarily publish milestone evidence — it is the cheapest way to be classified C3 rather than C4; expect diligence to demand verification-dated records
Economic-development organizations	Market regions with verified operating capacity and credible projects, not queue totals; queue gigawatts will correct, and credibility is the asset that survives
Journalists	Quote the class, the date, and the status: "a request in ERCOT's queue (April 2026)" is not "a data center being built"; use the taxonomy labels
Ratepayer advocates and residents	Participate in PUCT proceedings on cost allocation and curtailment implementation; ask your city and county which tax incentives are committed, to whom, and against what milestones
Texas agencies	Close the measurement gaps: a comprehensive data-center registry and mandatory water-use reporting would cost little and change the quality of every debate this report touches

9.1 Question checklist for communities and local governments

(Recommendation) A question list any resident, commissioner, or council member can use in a data-center conversation, keyed to the taxonomy. A project that cannot answer these is C4 at best.

1. What is the announced capacity, and is that figure an interconnection request, a contracted amount, or an operating measurement?

2. Who controls the site, under what instrument, and can we see the record?

3. Which ERCOT interconnection request number corresponds to this project, and what is its study status?

4. What water source, volume, and cooling technology are proposed — and is the disclosure in a permit or only in a press release?

5. What financing has closed, and what happens to our commitments if it does not?

6. What are the construction and permanent employment estimates, and what do comparable facilities actually employ?

7. Exactly what incentives are on the table, for how long, with what clawbacks keyed to which milestones?

8. What happens to the tax base, the water contract, and the equipment if the project is cancelled or curtailed long-term?

9. Is the load curtailable per SB 6, and does it have backup generation? What are the emissions and noise implications?

10. What is the expected energization date, and what evidence supports it?

10. Conclusion

Conclusion. The verified picture of Texas's AI infrastructure boom is a thin, real leading edge — about 9 GW observed or approved — in front of an enormous field of intentions: roughly 446 GW of interconnection requests, more than 320 GW of which have not yet submitted studies, inside a preliminary forecast that ERCOT itself says is likely to come in lower. Water measurement is too thin to settle the debate, and the state is rebuilding its forecasting and interconnection methods mid-stream — Batch Zero, forecast adjustments, and SB 6 implementation are all live. None of this is an argument for alarm or for complacency; it is an argument for evidence discipline. Edition 2 will publish the completed, verification-dated project database under the taxonomy and rules set out here, with independent technical review and a public correction log. Texas deserves a debate conducted on confirmed facts, typed claims, and honest ranges — and this report is built to hold that line.

References

Access dates: October 27, 2026. The final release will include archived snapshots where permitted.

1. ERCOT (2026, April 15). ERCOT Releases Preliminary Long-Term Load Forecast for Years 2026–2032 for PUCT Discussion. https://www.ercot.com/news/release/04152026-ercot-releases-preliminary

2. ERCOT filing to PUCT Project 58777 (2026, April 15). Preliminary Long-Term Load Forecast for Years 2026–2032. https://interchange.puc.texas.gov/Documents/58777_38_1622647.PDF

3. ERCOT Monthly, April 2026 (published May 12, 2026). https://www.ercot.com/files/docs/2026/05/13/ERCOT-Monthly-April-2026-FINAL.pdf

4. ERCOT Board of Directors, Item 9: Interconnection and Grid Analysis Update (2026, April 13). https://www.ercot.com/files/docs/2026/04/13/9-Interconnection-and-Grid-Analysis-Update.pdf

5. ERCOT Board of Directors, Item 12: CEO Update (2026, April 15). https://www.ercot.com/files/docs/2026/04/15/12-CEO-Update.pdf

6. ERCOT Market Notice M-B051826-01 (2026, May 18). Proposal to Adjust 2026 Long-Term Load Forecast. https://www.ercot.com/services/comm/mkt_notices/M-B051826-01

7. ERCOT (2026, June 18). PUCT Approves ERCOT's Batch Zero Process for Connecting Large Electricity Users While Protecting System Reliability for Texans. https://www.ercot.com/news/release/06182026-puct-approves-ercots

8. ERCOT Board of Directors, Item 8: System Planning and Weatherization Update (2026, September 7). https://www.ercot.com/files/docs/2026/09/07/8-System-Planning-and-Weatherization-Update.pdf

9. ERCOT (2026, April 1). ERCOT Update to the Senate Committee on Business and Commerce. https://www.ercot.com/files/docs/2026/04/01/ERCOT_LargeLoad_Update_April2026_B-C_-Hearing.pdf

10. Utility Dive (2026, April 16). ERCOT says Texas demand could quadruple but cautions forecast may be inflated. https://www.utilitydive.com/news/texas-demand-to-quadruple-by-2032-ercot-says-maybe-but-dont-bet-on-it/817698

11. Texas Senate Bill 6 (89th Legislature, 2025), enrolled text. https://capitol.texas.gov/tlodocs/89R/billtext/pdf/SB00006F.pdf

12. Texas Senate Research Center, Bill Analysis S.B. 6 (2025, August 5). https://capitol.texas.gov/tlodocs/89R/analysis/pdf/SB00006F.pdf

13. Utility Dive (2025, June 25). Texas law gives grid operator power to disconnect data centers during crisis. https://www.utilitydive.com/news/texas-law-gives-grid-operator-power-to-disconnect-data-centers-during-crisi/751587/

14. Shehabi, A., et al. (2024). 2024 United States Data Center Energy Usage Report. LBNL/DOE, December 2024. https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf

15. Lawrence Berkeley National Laboratory (2026, June). United States Data Center Energy Usage Report: 2025 Update. DOI: 10.71468/P1RP4F. https://eta-publications.lbl.gov/publications/united-states-data-center-energy-2025 (PDF: escholarship.org/content/qt33m6w3x0/qt33m6w3x0.pdf)

16. DOE (2024, December 20). DOE Releases New Report Evaluating Increase in Electricity Demand from Data Centers. https://www.energy.gov/articles/doe-releases-new-report-evaluating-increase-electricity-demand-data-centers

17. HARC (2026, January). Thirsty Data and the Lone Star State: The Impact of Data Center Growth on Texas's Water Supply. https://harcresearch.org/wp-content/uploads/2026/01/Thirsty-Data-Water-Use-and-The-Projected-Data-Center-Boom-in-Texas.pdf

18. UT News (2026, May 6). Data Centers Are Growing in Texas, But Big Questions Remain About Water Use (and UT white paper). https://news.utexas.edu/2026/05/06/data-centers-are-growing-in-texas-but-

big-questions-remain-about-water-use

19. Texas Tribune (2025, September 25). Data centers are thirsty for Texas' water, but amount is unclear. https://www.texastribune.org/2025/09/25/texas-data-center-water-use

20. Texas Scorecard (2025, October 13). Data Center Water Use Shrouded in Uncertainty. https://texasscorecard.com/state/data-center-water-use-shrouded-in-uncertainty/

21. Amarillo Globe-News (2025, September 26). Data centers are thirsty for Texas' water, but state planners are unclear on future total growth. https://www.amarillo.com/story/news/2025/09/26/data-centers-are-thirsty-for-texas-water-but-state-planners-are-unclear-on-future-total-growth/86368108007

22. Ceres (2026, August 25). Water Behind the Watts: The Hidden Risk of Powering Data Centers. https://www.ceres.org/resources/reports/water-behind-the-watts-the-hidden-risk-of-powering-data-centers

23. The Texan (2026, May 13). Texas Groundwater Conservation District Appeals to Legislature for Protection in Regulating Data Centers. https://thetexan.news/state/legislature/texas-groundwater-conservation-district-appeals-to-legislature-for-protection-in-regulating-data-centers/article_04662862-8632-4a7f-bdad-5795677811d6.html

24. Texas Tribune (2026, June 26). What you need to know about data centers in Texas. https://www.texastribune.org/2026/06/26/texas-data-center-guide-what-you-need-to-know

25. TPR (2025, August 15). Big Tech's big thirst — AI's demand for Texas water. https://www.tpr.org/environment/2025-08-15/big-techs-big-thirst-ais-demand-for-texas-water

26. Mayer Brown (2025). Important Texas Regulatory Updates for Data Centers (SB 6 summary). https://www.mayerbrown.com/en/insights/publications/2025/07/important-texas-regulatory-updates-for-data-centers

27. McGuireWoods (2025, June 6). 89th Texas Legislative Session Recap. https://www.mcguirewoods.com/client-resources/alerts/2025/6/89th-texas-legislative-session-recap-key-outcomes-for-power-generation-and-clean-energy-industry/

Note on source quality: ERCOT, PUCT, legislative, LBNL/DOE, and named-research-institution sources are primary. News summaries are used for context and for quoted utility statements, with primary sources preferred wherever available. Company-related descriptions (the Stargate Abilene footprint and specifications) are Company claims unless corroborated by filings or utility statements, and are typed as such wherever they appear.

Appendix A: Project taxonomy and confidence scoring

A.1 Milestone dimensions

Nine milestone dimensions, each scored 0–2 in the workbook (0 = no evidence; 1 = claimed with weak documentation; 2 = verified with primary documentation): site control; permits; interconnection status; contracted power; financing; equipment orders; construction activity; announced customers; expected-energization credibility.

A.2 Class-assignment rules

• **C1 Operating:** ERCOT Observed Energized status or equivalent operational evidence, verified within 90 days.

• **C2 Approved-not-energized:** ERCOT Approval-to-Energize status or an executed interconnection agreement, verified within 90 days.

• **C3 Credible:** site control (deed, lease, or option) and at least two of: permits filed or issued; interconnection study advanced beyond initial screening; contracted power; financing closed or disclosed in filings; major equipment orders evidenced.

• **C4 Early proposal:** announcement or queue entry without C3 evidence.

• **C5 Speculative or duplicate:** fails identity checks (same site or entity as another entry), no site control, or no status movement across two consecutive verifications at least 90 days apart.

A.3 De-duplication rules

Entries are matched on (site coordinates or parcel) OR (parent entity plus county) OR (interconnection request number). When two entries match any rule, the stronger-evidenced record absorbs the weaker, and the merge is logged in the database. Merges are reversible if new evidence distinguishes the entries.

Appendix B: Data dictionary for the project database

Table 15. Project database fields

Field	Definition
project_id	Stable identifier; retained across renames
project_name / parent_entity	As disclosed; parent entity where known
county / site_latlon	Texas county; site location at parcel level where public
interconnection_request_id	ERCOT request identifier where known
class (C1–C5)	Per Appendix A rules

Field	Definition
milestone scores	Nine dimensions, 0–2 each, with an evidence pointer per scored milestone
announced_mw / contracted_mw / energized_mw	Typed separately: claims versus contracted versus observed
expected_energization	As stated, with a credibility note
evidence_strength	Strong / Moderate / Weak / Claim-only (Appendix D)
last_verified_date	Date of last ERC verification
verification_source	The specific document or system checked
corrections_log_ref	Link to any correction entry

Appendix C: Methodology appendix (full protocol)

• **Source hierarchy:** ERCOT and PUCT primary records for grid quantities; statutes and analyses for legal content; LBNL/DOE for national energy; named research institutions for water; company disclosures only as typed claims; press for context and quoted officials, never as the sole basis for a quantity.

• **Verification cadence:** every active entry re-verified at least every 90 days; every fact carries its last verification date; stale entries are flagged, not silently retained.

• **De-duplication:** per Appendix A.3, with a reversible merge log.

• **Conflict handling:** where ERCOT systems and company claims conflict, both are recorded with dates; the conflict itself is a finding.

• **Confidential information:** described generically (for example, "an executed agreement exists") without publishing confidential terms.

• **Site visits (Edition 2):** photography from public rights-of-way only; observations typed as direct observation with date.

• **Statistical treatment:** realization rates are presented as intervals from historical interconnection data; no point prediction is published without an interval.

• **Interactive map (Edition 2):** shipped only if its interface meets the accessibility standard; a static accessible map and the full data table ship regardless.

Appendix D: Evidence-strength definitions

Table 16. Evidence-strength definitions

Strength	Definition
Strong	Two or more independent primary sources, or one primary source plus direct observation
Moderate	One primary source, current
Weak	Single secondary source or a dated primary source

Strength	Definition
Claim-only	Company or interested-party assertion, uncorroborated

Appendix E: Interview protocol

On-record or on-background designation chosen by the interviewee; quotes checked for accuracy; every statement typed as Confirmed fact, Company claim, or Stakeholder position; conflicting accounts presented side by side; no anonymous pejoratives; interviewees receive their quotes before publication for accuracy checking only — not for approval of context. Full question banks per stakeholder group are published with Edition 2.

Appendix F: Corrections and right-of-reply policy

1. Before final release, every organization materially discussed receives a factual-error notice covering the specific statements about it.

2. Organizations have 15 business days to identify factual errors, with sourcing.

3. Verified factual errors are corrected before publication; interpretations and conclusions remain ERC’s alone.

4. All corrections and the correction log are published on the canonical report page.

5. No organization receives advance access to conclusions, scenarios, or recommendations.

Appendix G: Stakeholder positions (attributed, neutral)

Table 17. Stakeholder positions as publicly stated

Stakeholder	Position (as publicly stated)	Source type
ERCOT	The preliminary forecast is likely higher than realized load growth; methodology refinement is underway	Source fact (official statements) [3]
PUCT	Approved Batch Zero to connect large loads in quantities and locations the Texas grid can reliably support	Source fact (official release) [7]
Groundwater conservation districts	Data centers could place disproportionate pressure on local aquifers; seeking clearer regulatory authority	Stakeholder position (press) [23]
TWDB (via spokesperson)	Self-reported water data is validated by cross-checks; no comprehensive data-center list exists	Source fact (official statements) [19][20]
Developers	Projects designed for water-efficient or closed-loop cooling; commitments vary by site	Company claims [24]
Community groups (Amarillo area)	Concerns about Ogallala Aquifer depletion from proposed facilities	Stakeholder position (press) [19]

(Positions are attributed and typed; the report takes no position on them.)

Appendix H: Scenario arithmetic

Inputs (with sources): R = 445.8 GW of large-load requests (ERCOT Monthly, April 2026) [3]; L = 5.9 GW observed energized (same source); A = 3.2 GW approved-not-operational (same source); 2026 projected summer peak 90.5–98 GW [10]; 2032 preliminary forecast 367.8 GW [1].

Formula: large-load addition = r × (R − L − A), where r is the realization rate. Illustrative 2032 peak = 2026 projected summer peak + addition. Sensitivity: results scale linearly in r; the 2026 peak range adds plus or minus 3.75 GW to every scenario.

Worked values: r = 5%: addition = 0.05 × 436.7 = 21.8 GW; peak = 113.3–119.8 GW. r = 15%: addition = 65.5 GW; peak = 156.0–163.5 GW. r = 30%: addition = 131.0 GW; peak = 221.5–229.0 GW. **Every scenario sits far below the preliminary 2032 forecast of 367.8 GW — the arithmetic makes ERCOT's own caveat concrete.**

Status: the realization rates are chosen to span the plausible range; Edition 2 replaces them with empirically fitted distributions from historical interconnection data (an Estimate with stated assumptions, never a forecast).

Appendix I: Starter project database (verified entries)

Scope statement. This starter table contains only entries ERC could verify against the cited sources by October 27, 2026. It is deliberately short: the taxonomy and rules (Appendices A–B) are the deliverable, and Edition 2 completes the census. Each entry shows its last verification date and evidence strength.

Table 18. Starter project database (verified entries only)

project_id	Name / entity	County	Class	Announced MW	Evidence summary	Strength	Last verified
TX-ABE-001	Stargate Abilene campus (Crusoe and related parties)	Taylor (near Abilene)	C3 (verify C2□C1 transition in Edition 2)	1,200 MW announced	Under construction per press and company statements; 940-acre site; initial 8M-gallon closed-loop fill per city utility statements	Moderate (Company claims plus utility statement; no ERCOT entry cited here)	2026-10-27
—	ERCOT aggregated Observed Energized large loads	Statewide	C1 (aggregate)	5,900 MW	ERCOT Monthly, April 2026	Strong (primary)	2026-10-27
—	ERCOT aggregated Approval-to-Energize, not operational	Statewide	C2 (aggregate)	3,200 MW	ERCOT Monthly, April 2026	Strong (primary)	2026-10-27

project_id	Name / entity	County	Class	Announced MW	Evidence summary	Strength	Last verified
—	Queue entries with no studies submitted	Statewide	C4/C5 (aggregate)	321,000 MW	ERCOT Monthly, April 2026	Strong (primary)	2026-10-27
—	Queue entries meeting Section 9.5 requirements	Statewide	C3 candidates (aggregate)	22,000 MW	ERCOT Monthly, April 2026	Strong (primary)	2026-10-27

(Named-project rows expand in Edition 2 as county, permit, PUCT, and disclosure records are verified. Aggregated rows are shown to demonstrate the database structure with fully verified values.)

Appendix J: How to read ERCOT and PUCT documents (a civic primer)

Purpose. Most public confusion in this debate comes from treating four different document types as one kind of number. This primer equips non-specialists.

• **Long-Term Load Forecast (LTLF):** ERCOT's planning forecast of future peak demand. The 2026 edition is preliminary, built from base economic forecasts plus utility-reported medium and large load intentions, and carries ERCOT's own caveat that it is likely higher than realized growth. Use it for planning envelopes, not predictions. Filed with the PUCT in Project 58777 (the Triennial Reliability Assessment).

• **Large Load Interconnection Queue:** ERCOT's tracking of requests from loads of 75 MW and above seeking connection. Entry is free of commitment; entries include duplicates and speculative projects. ERCOT publishes status categories (no studies submitted; under review; Section 9.5 requirements met) — the maturity distribution matters more than the total.

• **Board materials (Items 8, 9, 12):** ERCOT's regular Board updates contain the freshest official queue and planning statistics, with dates. These are the best citation for any current number.

• **ERCOT Monthly:** a public monthly summary — the most quotable single source for queue totals, observed energized loads, and forecast status.

• **PUCT project filings (interchange.puc.texas.gov):** formal filings with control numbers and item numbers. Project 58777 is the Triennial Reliability Assessment docket where the LTLF lives; Project 58481 covers large-load interconnection standards. Filings are the authoritative record when news summaries disagree.

• **Market notices (M- notices):** ERCOT's formal communications to market participants — for example, the May 2026 proposal to adjust the LTLF. They reveal methodology changes early.

• **Statutes and bill analyses (capitol.texas.gov):** SB 6 text is the law; the Senate Research Center analysis summarizes intent. Rulemakings at the PUCT implement it — the implementing details, not the bill, answer most operational questions.

• **What to record when quoting:** the document, its date, and its status (preliminary, proposed, approved). A number without those three attributes is not quotable under this report's standard.

Appendix K: Civic FAQ

Questions we expect from readers, answered with the report's evidence rules.

Is ERCOT forecasting 368 GW of data centers? No. The 367,790 MW figure is a preliminary total-demand forecast for 2032 that includes all load types; it is built largely from utility-reported large-load intentions; and ERCOT itself says it is likely higher than realized growth. Treat it as a planning envelope. [1][2][3]

So how much is actually running? ERCOT reported 5.9 GW of observed energized large loads plus 3.2 GW approved but not operational as of April 2026. Those are the directly observed layers. [3]

Why do headline gigawatt numbers keep changing? Because the queue grows monthly and the forecast methodology is being revised mid-stream (the May 2026 adjustment proposal; the December 2026 forecast delay). Quote the document, the date, and the status. [6][8]

Does a queue entry mean a data center is coming? No. It is a request. 321 GW of the queue has submitted no studies. Use the taxonomy: C4 or C5 until milestone evidence appears. [3]

Is the national growth number "15% of US electricity"? The credible federal-laboratory answer is a range: 9.5–15.3% by 2030, reference case 11.8%. Quoting only the high end misrepresents the study. [15]

How much water do data centers use in Texas? The honest answer is that measurement is incomplete: HARC estimates about 25 billion gallons per year today across cooling and generation; UT Austin bounds 2040 use at 3–9% of state water use depending on growth and cooling technology; and TWDB's survey response rate (18 of about 70) shows the data foundation is weak. [17][18][19]

Who pays for the grid upgrades? SB 6 requires large loads to contribute to interconnection-cost recovery and be curtailable; the 4CP allocation methodology is under reevaluation. The distributional answer is being written in PUCT proceedings right now. [11][12]

Can ERCOT turn data centers off in an emergency? Yes, within the SB 6 framework: non-critical large loads must be curtailable during firm load shed, and loads with backup generation can be directed to deploy it. [11][13]

Is ERC affiliated with any project or party in this debate? No. ERC holds no position, accepts no party funding for this series, and will give materially discussed organizations a factual-error opportunity without editorial control. [Disclosures; Appendix F]

What is “Section 9.5 requirements met”? It is ERCOT’s status for queue applicants that have met the planning-guide requirements — a meaningful maturity signal: 22 GW had reached it as of April 2026, versus 321 GW with no studies submitted. [3]

What is Batch Zero? The PUCT-approved ERCOT process (June 2026) that assesses large-load interconnection requests in capacity-limited batches — replacing a first-come queue that could not process 445 GW of requests. Its results determine which projects connect where, and when. [7]

Are the generation projects (storage, solar, gas) that would serve data centers more real than the data centers? The generation queue is even larger — about 453,562 MW of active requests, dominated by storage (177,642 MW) and solar (162,927 MW) — and subject to the same intention-versus-infrastructure distinction. Both queues must be filtered through the same taxonomy before anyone claims the power will exist. [9]

What single practice improves this debate most? Verification dating. If every journalist, official, and developer stamped their gigawatt figures with a document, a date, and a status, most of the confusion this report documents would disappear overnight. [Section 5.6]

Why does this report avoid one overall score or verdict? Because the underlying facts are distributed, fast-moving, and politically contested; a single verdict would misrepresent them. The report instead separates facts, claims, assumptions, positions, and conclusions, and gives every audience its own view. [Section 1.1]

Appendix M: The numbers card — every headline figure with its source and status

A one-page card of every number used in this debate, with its document, date, and status. Journalists and officials may quote any row verbatim with its provenance. All rows verified October 27, 2026.

Table 21. The numbers card

Number	What it is	Document and date	Status
367,790 MW	Preliminary total 2032 demand forecast, ERCOT region	Preliminary LTLF, filed with PUCT April 15, 2026 (Project 58777)	Preliminary; ERCOT says likely higher than realized growth [1][2][3]
278,003 MW	Preliminary total 2029 demand forecast	Same filing	Preliminary [2]
85,508 MW	ERCOT all-time peak demand (observed record)	Same filing	Observed fact [2]
90,500–98,000 MW	Projected summer 2026 peak	ERCOT comments, April 2026	Projection [10]
445.8 GW	Large-load interconnection requests	ERCOT Monthly, April 2026	Request tracking [3]
~87–89%	Share of queue that is data centers	Board Item 9 / June release	Tracking statistic [4][7]

Number	What it is	Document and date	Status
321 GW	Queue requests with no studies submitted	ERCOT Monthly, April 2026	Maturity statistic [3]
93.7 GW	Queue requests under ERCOT review	ERCOT Monthly, April 2026	Maturity statistic [3]
22 GW	Queue requests meeting Section 9.5 requirements	ERCOT Monthly, April 2026	Maturity statistic [3]
5.9 GW	Observed Energized large loads	ERCOT Monthly, April 2026	Observed fact [3]
3.2 GW	Approval to energize, not operational	ERCOT Monthly, April 2026	Status fact [3]
438,000+ MW	Large-load requests tracked at Batch Zero approval	ERCOT release, June 18, 2026	Approved process context [7]
453,562 MW	Active generation interconnection requests (2,008 requests)	ERCOT Senate briefing, April 1, 2026	Request tracking [9]
9.5–15.3% / 11.8%	US data-center share of electricity by 2030 (range / reference)	LBNL 2025 Update, June 2026, DOI 10.71468/P1RP4F	Scenario range [15]
176 TWh / 4.4%	US data-center electricity in 2023	LBNL 2024 Report / DOE	Observed estimate [14][16]
~25 Bgal/yr	Texas data-center water use today (cooling + generation)	HARC, January 2026	Study estimate [17]
up to 161 Bgal / 2.7%	Texas data-center water by 2030 (projection)	HARC, January 2026	Study projection [17]
3–9% by 2040	Texas water use possibly attributable to data centers	UT Austin white paper, May 2026	Study range [18]
18 of ~70	TWDB data-center water-survey responses, 2024	Texas Tribune, September 2025	Confirmed fact [19]

Usage rule. Quote the number with its document, date, and status — all three, every time. That rule, applied consistently, is most of the “reality check” this report exists to provide.

Appendix N: Accessibility conformance statement

Target. WCAG 2.2 Level AA plus PDF/UA as the internal ERC standard for public reports; this report serves civic audiences that include public entities subject to ADA Title II and Texas DIR accessibility requirements, so WCAG 2.1 Level AA is treated as a floor.

Implemented in this edition. Letter page size; 0.75-inch margins; 10.5 pt body text left-aligned (not justified); headings implemented as distinct paragraph styles in a two-to-three-level hierarchy (a fourth level is not used); consistent headers and footers carrying the report identifier, version, and page numbers; embedded TrueType typeface with Unicode-mapped searchable text; document language (en-US) declared in the PDF catalog; live hyperlinks with descriptive text and PDF bookmarks matching the table of contents; document title, author, subject, and keyword metadata set; tables with header rows and grid

borders (no layout-only tables), headers repeated across pages; figures with direct value labels so color is never the sole encoding, each with a source-and-denominator note; minimum 4.5:1 text contrast. The governing-qualification statement appears as real text on the cover, not as an image.

Deviations and planned remediation. This edition is generated from a programmatic layout engine that does not yet emit a complete PDF/UA tag tree; figures therefore carry their full data as adjacent text and tables rather than as alternative-text-tagged images. Before final release, the PDF will be remediated and validated per the ERC accessibility acceptance process: automated checker, PDF/UA validation (PAC or equivalent), manual tag-tree and reading-order inspection, keyboard and screen-reader sample testing, plain-text extraction review, and a recorded test log. The accessible HTML edition is the canonical reading experience; this PDF serves offline, print, and archival use. An alternate-format contact appears in the document-control section; readers may request HTML, large-print, or plain-text versions of any table or figure through that page.

Figures and tables. Figures 1 and 2 present every plotted value as text adjacent to its bar; their complete underlying data appear in Tables 5, 6, and the cited sources. All tables use real header rows, repeated across pages.

Glossary and acronyms

4CP — four coincident peak; the transmission cost-allocation methodology under reevaluation. **Batch Zero** — ERCOT’s PUCT-approved batch process for assessing large-load interconnection requests. **C1–C5** — this report’s project-confidence classes. **Curtailment (firm load shed)** — mandatory reduction of load during emergencies; SB 6 requires non-critical large loads to be curtailable. **DOI** — digital object identifier. **LTLF** — long-term load forecast. **Observed Energized** — ERCOT’s classification for metered, operating large loads. **PUCT** — Public Utility Commission of Texas. **Realization rate** — the share of requested capacity that actually energizes. **SB 6** — Texas Senate Bill 6 (89th Legislature, 2025). **Section 9.5** — the ERCOT planning-guide requirements for large-load interconnection applicants. **TDSP** — transmission and distribution service provider. **TI** — (not used; see report 4). **TWDB** — Texas Water Development Board. **TWh** — terawatt-hour.

Change log

Table 22. Version history

Version	Date	Change
1.0	2026-10-27	Initial public draft: verified anchor evidence; five-class taxonomy; scenario arithmetic; starter database; corrections and right-of-reply policy.
1.0a	2026-10-28	Production remediation: accessibility conformance statement added (Appendix N); embedded TrueType typefaces; document language (en-US) declared in the PDF catalog; accessibility pre-test record published; companion accessible HTML edition generated.

Version	Date	Change
2.0 (planned)	2027	Completed verification-dated project census; attributed interviews; independent technical review; interactive companion map if accessibility requirements are met.

Data and code statement

The dated project database (CSV/XLSX), data dictionary, scenario workbook, and source table will be published on the ERC canonical report page for ERC-RPT-2026-005 under CC BY 4.0, with collection scripts under an OSI-approved license. The database is versioned: every edition freezes a snapshot, every entry carries its last verification date, and every change between editions is logged. An interactive companion map is planned for Edition 2 if accessibility requirements can be met for the map interface; a static accessible map and the full data table will ship regardless.

Corrections. Corrections, replacement versions, the database, and source tables will be posted on the canonical ERC report page identified in the document-control section. Readers — including materially discussed organizations — may submit factual corrections through that page under the Appendix F policy.